

A study on machine learning prediction of bio-oil yield from biomass and plastic Co-pyrolysis

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ABSTRACT

The co-pyrolysis of biomass and plastics can effectively enhance the quality of bio-oil. The application of machine learning techniques to predict bio-oil yield helps optimize the production of co-pyrolysis bio-oil. This study develops machine learning models for predicting bio-oil yield based on Deep Neural Networks (DNN) and Lightweight Gradient Boosting Machines. The study innovatively integrates the pyrolysis data of the three major components of biomass (cellulose, hemicellulose, and lignin), both individually and in mixtures, into the co-pyrolysis prediction model, overcoming the limitations of traditional studies that focus solely on the overall characteristics of biomass. The results show that the DNN model outperforms others, with the incorporation of biomass component data significantly improving the prediction accuracy of co-pyrolysis bio-oil yield, increasing the R^2 from 0.817 to 0.931, with an average absolute error of 3.583 and a root mean square error of 4.573. Additionally, analyses using Shapley additive explanations and Pearson correlation coefficients reveal significant changes in the feature importance ranking of the model, dynamically unveiling the impact mechanism of data expansion on feature weights. For the first time, the synergistic effect of plastic proportion and hydrogen content is explicitly identified. This research contributes to a deeper understanding of biomass pyrolysis mechanisms, thereby enhancing the economic value of co-pyrolysis bio-oil.

1. Introduction

The increasingly severe environmental issues, the growing population, and economic development have all driven the demand for renewable energy to mitigate the consumption of non-renewable resources. Biomass is now widely recognized as a major source of renewable energy worldwide [1]. Through thermochemical methods, biomass can be converted into more valuable by-products, including condensable volatile products (bio-oil), non-condensable gases, and biochar [2,3]. However, bio-oil exhibits characteristics such as high viscosity, high moisture and ash content, low calorific value, instability, and high corrosiveness [4]. The higher heating value of bio-oil typically ranges from 15 to 20 MJ/kg, which is only 40–50 % of that of conventional petroleum fuels (42–45 MJ/kg) [5]. Thus, the potential for bio-oil to directly substitute petroleum fuels and chemical feedstocks is limited. Consequently, further research on bio-oil is essential for providing liquid products that can be used as liquid fuels or chemical feedstocks in various applications [6].

In the past few decades, plastic production has increased twenty-

fold. Furthermore, this trend is expected to continue, with annual plastic production projected to exceed 500 million tons by 2050 [7]. As a widely available material, plastics can be effectively converted into hydrocarbon fuels through pyrolysis [8], with minimal environmental pollution [9,10]. The high volatile matter and low ash content of plastic waste indicate that they can serve as primary products generating useable oils and gases during pyrolysis [11,12]. Uzoejinwa et al. [13] found that co-pyrolysis of biomass and plastics significantly reduced the oxygen content in the resulting bio-oil while increasing the selectivity for aromatic compounds. This suggests that the co-pyrolysis of biomass and plastics contributes to enhancing the quality of bio-oil, further optimizing resource recovery and utilization.

Numerous factors influence the yield and quality of bio-oil produced from the co-pyrolysis of biomass and plastics, including feedstock properties, pyrolysis conditions, and feedstock ratios. Research by Dorado et al. [14] indicates that feedstock ratio is a key parameter during co-pyrolysis, significantly impacting the yield and quality of the final products. Currently, research has predominantly focused on feedstock ratio, pyrolysis temperature, and pyrolysis time; however, the heating

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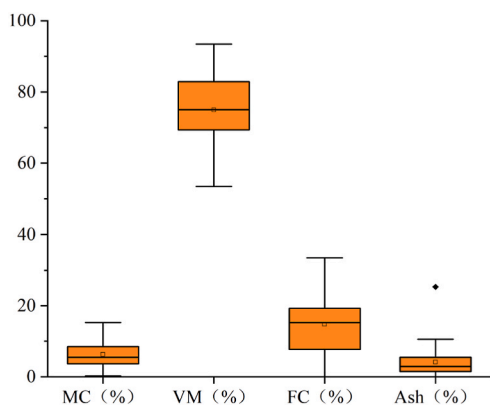
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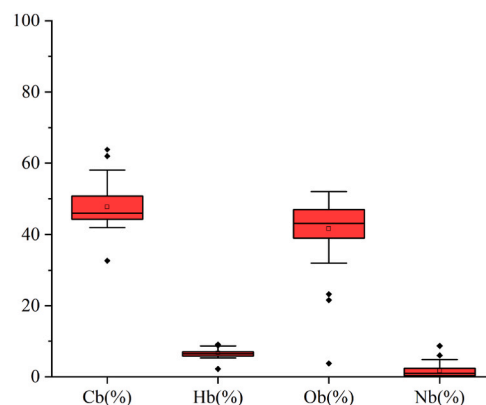
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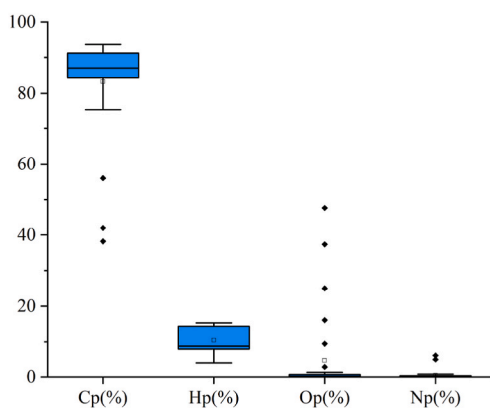
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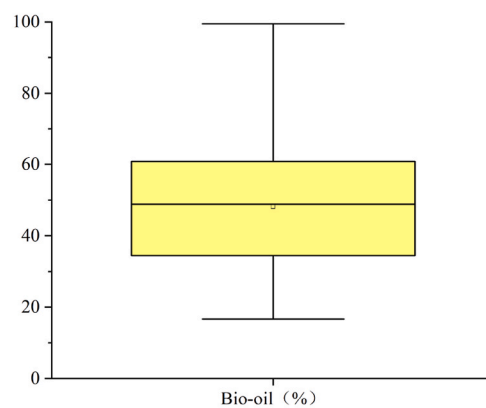
a) Biomass industry analysis



b) Biomass elemental analysis



c) Plastic elemental analysis



d) Bio-oil yield

Fig. 1. Box plot of biomass and plastic co-pyrolysis bio-oil yield dataset.

rate and the compositional makeup of biomass and plastics should not be overlooked [13]. Major components found in various types of biomass include cellulose, hemicellulose, and lignin. Worasuwannarak et al. [15] investigated the pyrolytic behavior of these three components and their mixtures using thermogravimetric-mass spectrometry (TG - MS) techniques, revealing significant interactions between cellulose and lignin that inhibit the formation of liquid products. Therefore, a certain degree of synergy exists among the three-component biomass data, significantly affecting the yield and properties of pyrolysis products.

With the development of artificial intelligence, many novel and advanced methods have been applied to various fields [16]. In current research, technologies such as Deep Neural Networks (DNN), Light Gradient Boosting Machines (LightGBM), Support Vector Machines (SVM), and Random Forests (RF) have been successfully applied to biomass pyrolysis, significantly reducing the cost of traditional experiments [17–19]. Regarding the research on the co-pyrolysis of biomass and plastics, Khandelwal et al. [20] and Bhushan et al. [21] conducted systematic reviews, both of which pointed out that machine learning techniques can effectively predict the bio-oil yield and techno-economic feasibility, providing data-driven solutions for process optimization. Predicting the bio-oil yield based on raw material characteristics, reaction conditions, and plastic content in the feedstock is currently quite

common. However, different models yield varying prediction accuracies, and the results regarding the influence of features also differ. Alabdrabalnabi et al. [22] and Xu et al. [23] used DNN and LightGBM models to predict the bio-oil yield, with prediction accuracies (R^2) ranging from 0.86 to 0.96, discovering that reaction conditions and raw material ratios have a significant impact on the bio-oil yield. Prasertpong et al. [24] and Potnuri et al. [25] used eXtreme Gradient Boosting (XGBoost) and SVM models to predict the bio-oil yield, with R^2 between 0.90 and 0.96, revealing the synergistic effect in the co-pyrolysis of biomass and plastics. Tang et al. [26], Yang et al. [27], and Ullah et al. [28], based on the characteristics of biomass and pyrolysis conditions, used RF and other models to predict the bio-oil yield, with R^2 between 0.85 and 0.95 and found that pyrolysis conditions, especially the heating rate, have the greatest impact on bio-oil yield predictions, with a relative contribution higher than the biomass characteristics. Meanwhile, selecting or expanding model features is essential for improving prediction accuracy. From the above, it can be observed that improving prediction performance by considering a broader range of input features can effectively optimize process parameters, thereby enhancing both the yield and quality of bio-oil [29, 30]. However, existing studies on predicting bio-oil yield from biomass and plastic co-pyrolysis primarily focus on different algorithms and the

selection of input features. Therefore, exploring new methods to significantly optimize the prediction model for co-pyrolysis bio-oil yield, improving prediction accuracy, and deepening the understanding of the potential mechanisms of co-pyrolysis is crucial.

This study developed a machine learning model based on DNN and LightGBM to predict the bio-oil yield during the co-pyrolysis of biomass and plastic. The model is the first to integrate the individual and combined pyrolysis data of the three major components of biomass (cellulose, hemicellulose, and lignin) into the co-pyrolysis prediction, utilizing expanded data for improved prediction. The research investigates whether the inclusion of three-component data affects the performance and accuracy of the prediction model, and employs Shapley Additive Explanations (SHAP) value analysis to explore the impact of the inclusion of three-component data on the significance of input features, as well as their synergistic or inhibitory effects.

2. Research methods

2.1. Feature selection for model input

Akhtar and Amin [31] provided a detailed discussion on the parameters affecting bio-oil yield, concluding that the biomass pyrolysis process is influenced by many parameters, including biomass type, temperature, heating rate, and reaction time. In co-pyrolysis, unlike conventional pyrolysis, the ratio of feedstock also affects the yield of pyrolysis products. Sharypov et al. [32] studied the co-pyrolysis of wood biomass and synthetic polymer mixtures, determining that the ratio of biomass to plastic in the feedstock is crucial for bio-oil yield. Abnisa et al. [33] reached a similar conclusion in their investigation of the co-pyrolysis of palm kernel shells and polystyrene waste, exploring the effects of temperature, feedstock ratio, and reaction time on co-pyrolysis, revealing that the feedstock ratio is the most significant variable affecting liquid product yield.

Since the feedstock ratio is the most significant variable affecting the liquid product yield, the factors influencing the bio-oil yield from biomass and plastic co-pyrolysis primarily stem from the basic properties of both feedstocks and the pyrolysis conditions. This study includes an industrial analysis of biomass, comprising moisture content (MC), volatile matter (VM), fixed carbon (FC), and ash content (Ash), along with elemental analyses of biomass and plastic, which include carbon (C), hydrogen (H), oxygen (O), and nitrogen (N) contents as input features. The elemental analysis of biomass is denoted with the suffix 'b' (Cb, Hb, Ob, Nb), while that of plastic is denoted with 'p' (Cp, Hp, Op, Np). The pyrolysis conditions selected include reaction time (RT), pyrolysis temperature (PT), and heating rate (HR), with the proportion of plastic in the feedstock (PR) also included as one of the input features. The output feature is the bio-oil yield (Bio-oil).

2.2. Data sources and experimental design

A comprehensive literature review was conducted to gather data on the co-pyrolysis of biomass and plastic, resulting in a dataset consisting of 122 data points representing various types of biomass, including sugarcane bagasse, almond shells, and grape seeds. The dataset encompasses a wide variety of biomass types, with some data originating from experiments and others sourced from published literature. The sources of the data are detailed in the supplementary materials, which are provided at the end of the article. C Couhert et al. [34] explored the possibility of replacing biomass pyrolysis with the pyrolysis of individual components such as cellulose, hemicellulose, and lignin, and found that there are interactions between these components. Subsequent research has validated that the main components of biomass interact, indicating possible synergistic or inhibitory effects [5]. Due to the complexity of biomass, collecting large sample datasets is often difficult, and training models on limited data may lead to the omission of these interactions. To address this, data from the individual and mixed

pyrolysis of the three major components (cellulose, hemicellulose, and lignin) were included to improve the model's capacity to capture these potential responses.

A boxplot was constructed to visualize the sample data obtained from biomass industrial analysis, elemental analysis, and plastic elemental analysis, as well as the bio-oil yield data, in order to identify anomalies and facilitate the recognition of outliers within these datasets.

From Fig. 1, it is evident that the elemental analysis of biomass and plastics exhibits numerous anomalies, which are considered to be influenced by the addition of three-component pyrolysis data. While the pyrolysis characteristics of biomass are affected by three primary components, the interactions among these components significantly influence the product distribution, particularly for products containing macromolecules such as bio-oil [35].

Analysis of Fig. 1a indicates that the dataset for biomass exhibits a higher content of VM and FC, with VM content being the most prominent, primarily concentrated between 67 % and 85 %, while FC is mainly between 10 % and 20 %. The MC and Ash content are relatively low, each below 10 %. Fig. 1b reveals a substantial presence of C and O elements in biomass, predominantly within the range of 40 %–50 %, while the contents of H and N are comparatively low and narrow in range. In Fig. 1c, it can be observed that plastics contain a significant amount of C, which differs markedly from other elements, reaching 85 %–95 %. The average H content is 10 %, showing no anomalous points. However, there are numerous anomalies in the O content, indicating considerable variability among different plastics. Fig. 1d shows that the bio-oil yield is concentrated between 30 % and 60 %, with no anomalous data present. In summary, the dataset does not exhibit significant anomalies, and due to the similar fundamental characteristics observed between different biomass and plastics, no data exclusion is warranted.

This study introduces three-component analysis data into the machine-learning context of biomass and plastic co-pyrolysis for the first time. Three sample groups were designed for experimentation: Sample A contains only the co-pyrolysis data of biomass and plastic; Based on sample A, the co-pyrolysis data of individual biomass components (cellulose, hemicellulose, and lignin) with plastic were included, designated as sample B, so as to compare with sample A to explore the effect of introducing individual biomass components on the co-pyrolysis; Building upon sample B, the co-pyrolysis data of biomass component mixtures (cellulose/hemicellulose, cellulose/lignin, hemicellulose/lignin, and cellulose/hemicellulose/lignin) with plastic, mixed in equal proportions, were incorporated, designated as sample C, in order to further explore the impact of the co-pyrolysis of mixed biomass components on co-pyrolysis, and to compare it with sample B. The models of the three sample groups were trained, and the training results were compared to conclude. Segregating of the experimental samples can evaluate how different combinations of biomass components affect the overall process. These three groups enable structured comparisons and provide a deeper understanding of how the introduction of individual or mixed biomass components alters the co-pyrolysis behavior, which is crucial for process optimization.

2.3. Model selection and hyperparameter tuning

Due to the complexity of biomass, this study employs a small sample dataset and selects DNN and LightGBM models for analytical purposes. The application of DNNs in small datasets within materials research is becoming increasingly significant. Compared to traditional machine learning methods (such as Support Vector Networks (SVN), RF [36,37], and SVM [38]), DNNs can offer greater accuracy and superior generalization capabilities when dealing with smaller datasets [39]. LightGBM, an ensemble learning method based on gradient-boosted trees, is noted for its high training efficiency, rapid speed, and superior accuracy [40]. The models were developed using Python version 3.8, with the DNN model implemented via libraries such as Sklearn, TensorFlow, and PyTorch, while the LightGBM model was constructed using its

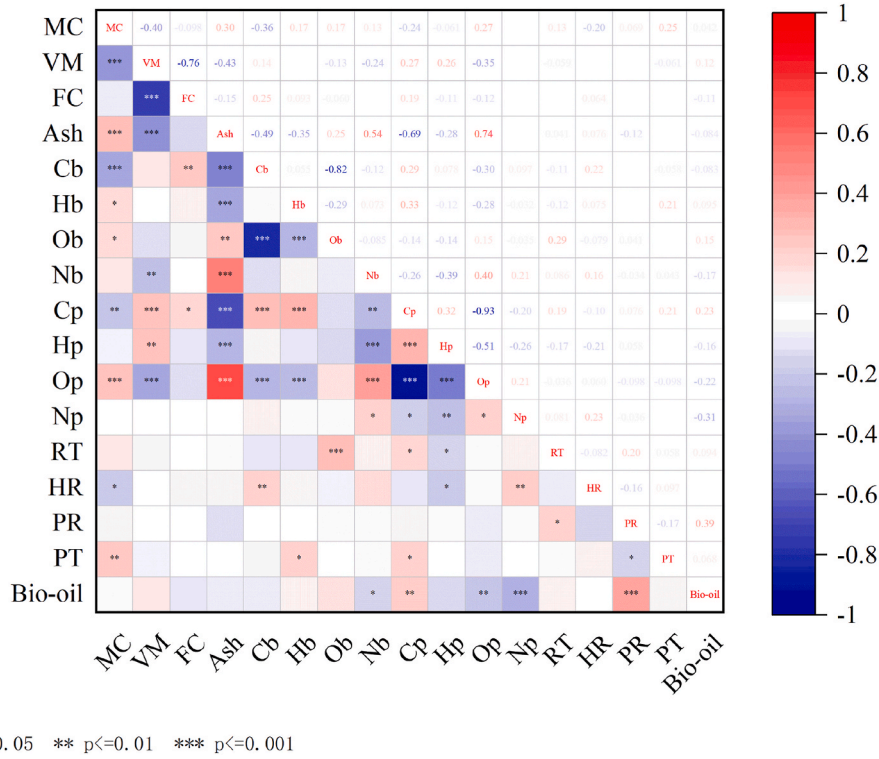


Fig. 2. Correlation analysis of biomass and plastic Co-pyrolysis oil yield dataset.

proprietary library.

The DNN model, also known as a multilayer perceptron, comprises an input layer, hidden layers, and an output layer, with full connectivity between layers. The Rectified Linear Unit (ReLU) activation function was chosen for its superior performance in prediction accuracy, particularly due to its ability to mitigate the vanishing gradient problem common with activation functions like Logistic Sigmoid Function or Hyperbolic Tangent Function, as well as its facilitation of faster convergence during training, which is crucial for deep learning models [41]. The loss function optimization was compared among Stochastic Gradient Descent (SGD), Adaptive Moment Estimation (Adam), and Nesterov-accelerated Adaptive Moment Estimation (Nadam), with Adam being ultimately selected due to its superior effectiveness in terms of prediction accuracy, loss reduction, faster convergence, and improved stability, particularly when handling sparse gradients, in comparison to SGD and Nadam. Given the substantial variance among input features, feature normalization was conducted, with related parameters preserved for subsequent inverse normalization to ensure the accuracy of output results. Given the substantial variance among the input features, feature normalization was performed using MinMaxScaler, and the scaling coefficients were saved in pkl format using the dump function. These coefficients were then loaded from the pkl file and used for inverse normalization. The hidden layers were configured as two layers, with the first containing 64 neurons and the second containing 32 neurons, where the output a_j^l of the j th neuron in the l th layer is:

$$a_j^l = \sigma(z_j^l) = \sigma\left(\sum_{k=1}^m w_{jk}^l a_k^{l-1} + b_j^l\right) \quad (1)$$

Where: w_{jk}^l is the weight of the j th neural node connecting the l th layer, a_k^{l-1} is the l -1st layer output, and b_j^l is the bias term connecting the j th neural node of the l th layer.

In equation (1), l is set to 3 and j is set to 1 for the output of this model. Additionally, the batch size is set to 16, the training period is set to 400, and the learning rate is set to 0.25. The L1 and L2 regularization coefficients are set to 0.1 and 0.01, respectively.

The LightGBM model, built upon traditional Gradient Boosting Decision Trees (GBDT), incorporates Gradient-based One-Sided Sampling (GOSS) and Exclusive Feature Bundling (EFB). A histogram-based approach is used for data processing and feature storage. A maximum depth limit was imposed on the leaf-wise growth strategy to avoid overfitting resulting from excessive tree splits [42]. In this study, after continuous comparison and adjustment, the optimal parameter combination was selected to achieve the highest prediction accuracy. The specific settings of some parameters are as follows: max_depth = 4, min_split_gain = 0.01, min_child_weight = 8, random_state = 16, and early_stopping_rounds = 10.

2.4. Model training and evaluation metrics

The dataset was randomly divided into training and testing sets at a ratio of 3:1, with the training set utilized for model construction and the testing set for accuracy evaluation. In the hyperparameter tuning phase, cross-validation was employed, using Mean Absolute Error (MAE) as the basis for selecting optimal hyperparameters. Once hyperparameters were determined, the model was retrained, and 25 % of the data was used for testing, and evaluating model performance using the R^2 and Root Mean Square Error (RMSE). Variability in training sets may lead to differences in sample distribution and feature tendencies, making the use of a single training set potentially limiting for model generalization [43]. To enhance the model's robustness and reproducibility against input perturbations, multiple random seeds were utilized to generate different training and testing sets, typically involving around 30 trials for reliable performance assessment.

R^2 represents the ratio of the regression sum of squares to the total sum of squares, gauging the model's fit quality with a range between 0 and 1, where values closer to 1 indicate better fitting. However, it should be complemented with other metrics to avoid overfitting. MAE measures the average absolute difference between predicted and actual values, making it a widely used metric for assessing regression model performance. Smaller MAE values indicate higher accuracy, and unlike RMSE, it is less influenced by outliers [44]. RMSE, on the other hand, is a

Table 1
Performance of the LightGBM model.

Scenarios		Training			Testing		
		a	b	c	a	b	c
R^2	1	0.889	0.897	0.880	0.821	0.8702	0.913
	2	0.861	0.828	0.909	0.852	0.878	0.883
	3	0.874	0.831	0.822	0.854	0.870	0.851
	AVG	0.875	0.852	0.870	0.842	0.873	0.882
MAE	1	4.049	3.764	4.539	4.786	5.065	3.520
	2	4.442	4.482	3.624	5.004	5.340	4.411
	3	4.166	5.416	5.600	5.531	4.247	3.951
	AVG	4.219	4.554	4.588	5.107	4.884	3.961
RMSE	1	5.723	5.321	5.939	6.363	6.262	4.517
	2	6.163	5.863	5.012	6.471	7.630	5.859
	3	5.692	7.017	7.278	6.957	5.679	5.804
	AVG	5.859	6.067	6.076	6.597	6.524	5.393

standard metric for evaluating models that predict numeric outcomes [45,46]. It quantifies the magnitude of discrepancies between predicted and actual values, with lower values indicating a better fit. The formulas for R^2 , MAE, and RMSE are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2} \quad (2)$$

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (3)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2} \quad (4)$$

Where y_i represents the true value, $\hat{y}_i$ represents the predicted value, $\bar{y}$ represents the average value, and m is the number of samples.

3. Results and discussion

3.1. Data Correlation Analysis

The Pearson Correlation Coefficient (PCC) is commonly used to assess the degree of correlation between variables, with values ranging from -1 to 1 . A negative value indicates a negative correlation, a positive value indicates a positive correlation and a correlation coefficient of 0 signifies no correlation [47].

From Fig. 2, it is evident that three features show a significant correlation with the output bio-oil, with $p \leq 0.001$: Op, Np, and PR. Among these, PR demonstrates a positive correlation with a correlation coefficient of 0.45 , which is the highest among all feature correlations, indicating that PR has the most significant impact on bio-oil yield. This finding is consistent with the conclusions drawn by Alabdrabalnabi et al. [22]. In contrast, Np and Op exhibit negative correlations with bio-oil, with correlation coefficients of -0.38 and -0.32 , respectively, suggesting that both factors also have important effects on bio-oil yield, ranking as the second and third most influential factors.

The features VM, Nb, and Cp show a significant correlation with $p \leq 0.01$. Both VM and Cp present positive correlations, with correlation coefficients of 0.24 and 0.27 , while Nb shows a negative correlation with a coefficient of -0.25 . The feature PT is significant at $p \leq 0.1$. Although other features exhibit relatively low correlations with oil yield, notable correlations among these features exist [31]. Given that the significant features encompass the fundamental properties of biomass and plastic as well as pyrolysis conditions, their potential impact warrants that no features be excluded from further analysis.

3.2. Prediction accuracy analysis

Table 1 records the training results for three sample groups. For both

Table 2
Performance of the DNN model.

Scenarios		Training			Testing		
		a	b	c	a	b	c
R^2	1	0.935	0.925	0.901	0.811	0.905	0.929
	2	0.907	0.915	0.913	0.822	0.857	0.931
	3	0.925	0.898	0.930	0.817	0.860	0.932
	AVG	0.922	0.913	0.915	0.817	0.874	0.931
MAE	1	3.309	3.588	3.476	4.783	2.794	3.661
	2	3.683	3.906	3.215	4.418	3.300	3.571
	3	3.683	3.962	3.156	4.765	3.174	3.516
	AVG	3.558	3.819	3.282	4.655	3.089	3.583
RMSE	1	4.374	4.925	5.208	6.316	3.419	4.630
	2	5.230	5.221	4.873	6.128	4.210	4.575
	3	4.706	5.729	4.380	6.217	4.164	4.514
	AVG	4.770	5.292	4.820	6.220	3.931	4.573

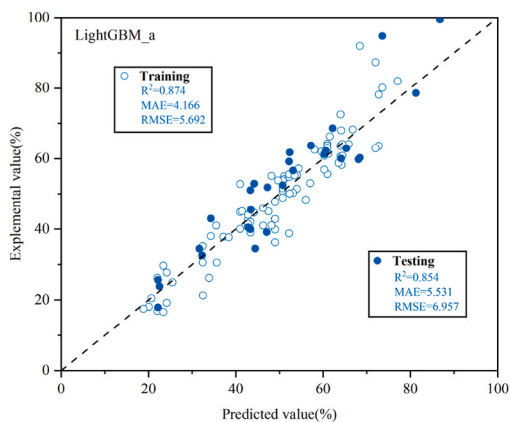
the LightGBM and DNN models, 30 training repetitions were performed, and the three best results were recorded in the table. Due to the interference of randomness and noise, the model's output may not accurately reflect its true performance. To enhance the scientific rigor and reliability of the experimental results, it is crucial to avoid evaluating based solely on a single optimal outcome. Instead, taking the average of the three experimental results (AVG) is recommended to ensure a thorough assessment. Furthermore, attention should be given to a comprehensive set of indicators to evaluate the model holistically [48].

Based on the comparative analysis of the test dataset, it was observed that the inclusion of three-component analytical data resulted in an increase of 0.04 in the R^2 of the LightGBM model, while the R^2 of the DNN model improved by 0.114 . Additionally, both the MAE and the RMSE showed a noticeable decrease, indicating an overall enhancement in model accuracy.

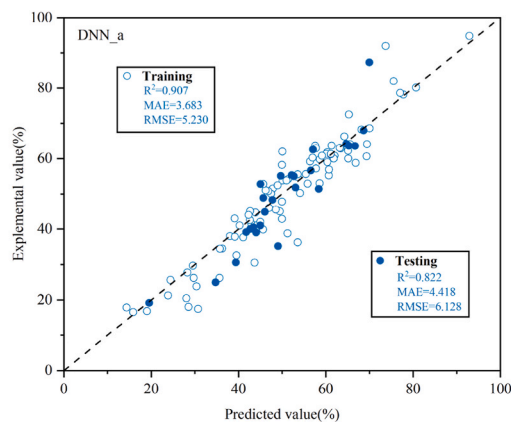
As shown in Table 1, the R^2 value for LightGBM_a is 0.842 . Upon the incorporation of the three-component data analyzed separately with the plastic co-pyrolysis data, the R^2 increased to 0.873 , an improvement of 0.031 . Further inclusion of the three-component mixed co-pyrolysis data raised the R^2 to 0.882 , with the MAE decreasing from an initial value of 5.107 to 3.961 , and the RMSE reducing to 5.393 . After incorporating the three-component data, the R^2 reached 0.886 , which represents an improvement in model performance compared to the R^2 of 0.86 reported by Xu et al. [23], demonstrating that the inclusion of the three-component data effectively enhanced the performance of the LightGBM model. Notably, LightGBM_c exhibited the lowest MAE and RMSE among the three models, which also further indicates that the integration of three-component data effectively enhances the accuracy of the LightGBM model.

In Table 2, regarding the DNN model, DNN_b showed a significant improvement compared to DNN_a, with the R^2 rising from 0.817 to 0.874 , while the MAE decreased to 3.089 and the RMSE reduced to 3.931 . This indicates a substantial enhancement in model performance. However, after the addition of the three-component mixed pyrolysis data, DNN_c achieved an R^2 of 0.931 . There is a significant improvement in the R^2 value compared to DNN_b, demonstrating an R^2 of 0.93 , which exceeds the 0.92 reported by Tang et al. [30] for their bio-oil model. However, compared to DNN_b, both the MAE and RMSE increased to 3.583 and 4.573 , respectively.

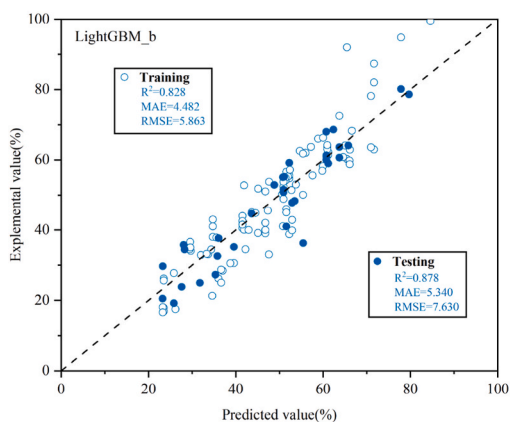
According to previous research, R^2 provides more informative and authentic insights than Symmetric Mean Absolute Percentage Error (SMAPE) and does not carry the interpretability limitations associated with Mean Squared Error (MSE), RMSE, MAE, and Mean Absolute Percentage Error (MAPE) [49]. Therefore, a comprehensive comparison suggests that DNN_c exhibits superior model accuracy. The degradation in MAE and RMSE performance following the integration of three-component mixed pyrolysis data may be attributed to the intrinsic nature of neural networks as "black box" models. When the training sample size is insufficient, the potential reaction mechanisms between



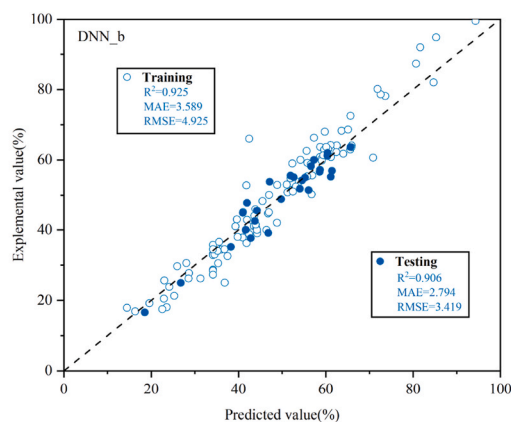
a) LightGBM_a



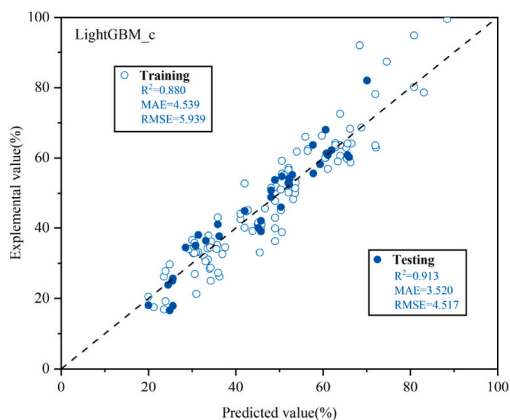
b) DNN_a



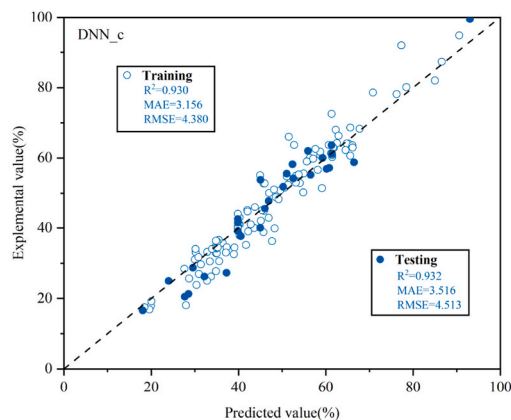
c) LightGBM_b



d) DNN_b



e) LightGBM_c



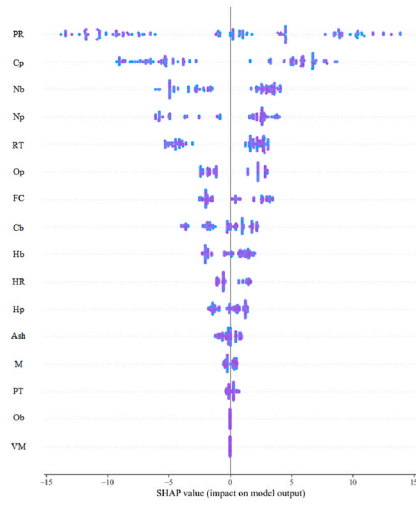
f) DNN_c

Fig. 3. Comparison of the predicted value of LightGBM and DNN models with experimental values.

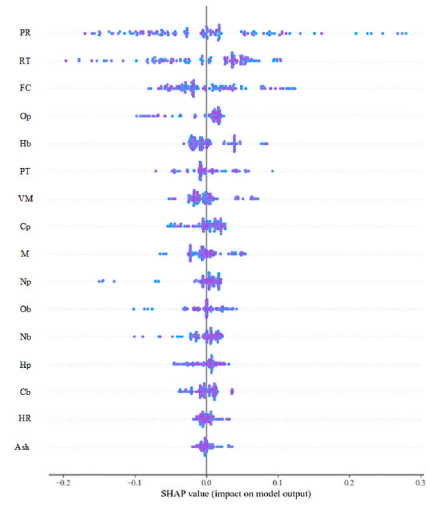
the three major biomass components and plastic co-pyrolysis may not be effectively captured by the model. This could result in certain nodes possessing lower weights, making it challenging for the model to identify these crucial reaction mechanisms during prediction. However, the incorporation of three-component and plastic co-pyrolysis data

significantly enhanced the relevant weights, allowing the model to better focus on these latent reaction mechanisms [50]. Consequently, the model may have experienced slight overfitting when addressing complex reactions, leading to a decline in performance.

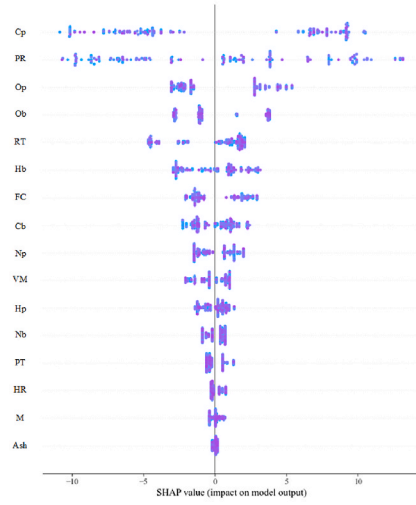
Fig. 3 illustrates that the dashed line represents the ideal scenario



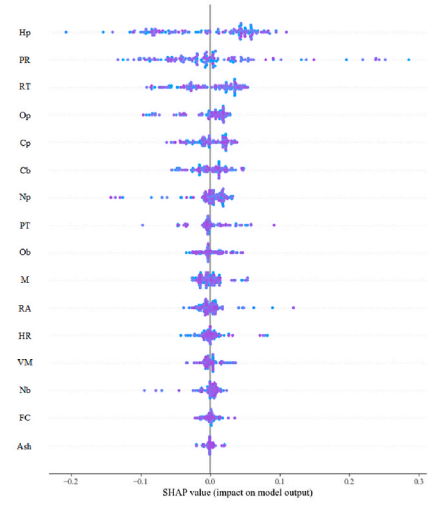
a) LightGBM_a



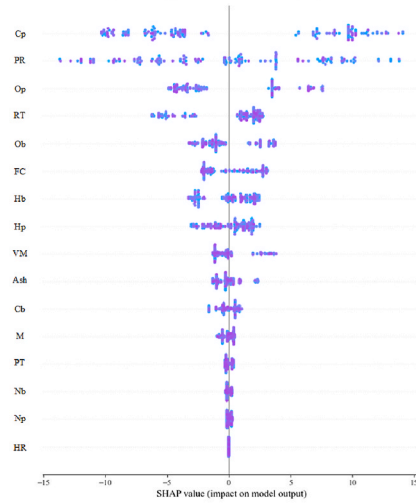
b) DNN_a



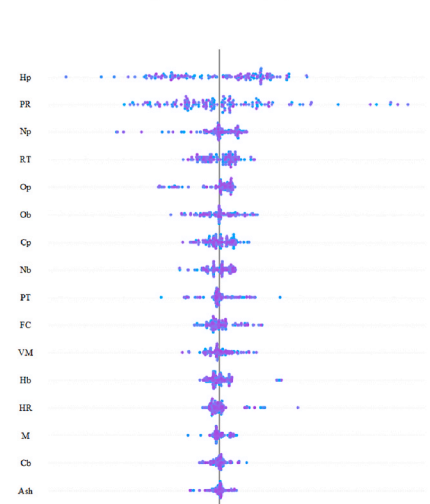
c) LightGBM_b



d) DNN_b



e) LightGBM_c



f) DNN_c

Fig. 4. LightGBM model and DNN model SHAP value analysis.

where experimental data perfectly aligns with the model predictions. The closer the data points are to the dashed line, the better the predictive performance of the model. Overall observation indicates that the scatter plot for the DNN model is more concentrated around the line $y = x$ compared to that of the LightGBM model, suggesting that the DNN model outperforms the LightGBM model in predicting bio-oil yield in this study. This finding is consistent with the results of Yang et al. [27].

A comparative analysis of the LightGBM scatter plots reveals that the performance of LightGBM_c is superior to that of LightGBM_a and LightGBM_b (see Fig. 4). In analyzing the DNN scatter plots, it is evident that DNN_b and DNN_c exhibit a more concentrated and orderly distribution compared to DNN_a, and both outperform the LightGBM models. Given that the scatter plots for DNN_b and DNN_c visually appear similar, and considering the metrics R^2 , MAE, and RMSE, it can be concluded that DNN_c possesses the highest accuracy and can be regarded as the optimal model for predicting bio-oil yield.

In summary, the comparative analysis indicates that the DNN model demonstrates superior performance in predicting the bio-oil yield from the co-pyrolysis of biomass and plastic, with DNN_c identified as the best predictive model.

3.3. SHAP value analysis

SHAP values represent the contribution of each feature to the model's predictions, where positive values indicate an increase in predicted value, and negative values indicate a decrease. By assessing the stacking degree and coverage area of SHAP values, the features can be ranked according to their impact on predictions [51]. Therefore, this study conducts a SHAP value analysis to explore the important features influencing the bio-oil yield from the co-pyrolysis of biomass and plastic.

The comparison in the figure reveals that incorporating three-component pyrolysis data into the samples alters the influence of various input features on bio-oil yield. An analysis of SHAP values conducted on the trained model indicates that the PR feature has both negative and positive SHAP values with coverage points, and its accumulation is less than that of other features. This suggests that the PR feature is the most significant factor influencing bio-oil yield during co-pyrolysis, which aligns with the findings of Prasertpong et al. [24], who also identified PR as the primary factor affecting bio-oil yield, noting that higher plastic content in the feedstock leads to increased liquid yield from pyrolysis [52]. The co-pyrolysis of biomass and plastics occurs in two stages. The first stage primarily involves the decomposition of biomass, with minimal influence on the decomposition of plastic materials. The second stage is dominated by the thermal degradation of plastics and lignin-derived biomass. Due to its thermal stability, the lignin component of biomass undergoes significant decomposition during this phase. The synergistic effect, resulting from the interaction between biomass and plastics, is significant within this temperature range. This synergy notably affects the yield of bio-oil, with the intensity of the peaks in both the first and second stages depending on the blending ratio of the mixture, thereby exerting a considerable impact on the bio-oil yield [53].

Given that the DNN model performs better than the LightGBM model in predicting bio-oil yield, further analysis of the DNN model's SHAP values is warranted. According to figure d), without the inclusion of three-component data, the PR feature stands out as the most significant. However, after incorporating the three-component data, figures e) and f) reveal that the SHAP values of the Hp feature show broader coverage than those of the PR feature, indicating that the hydrogen content in plastics significantly impacts bio-oil yield. Furthermore, the SHAP value analysis by Alabdrabalnabi et al. [22] also demonstrates that the Hp feature ranks highly, underscoring its significant influence on the bio-oil yield. Studies have shown that the unique elemental composition of plastics and biomass allows the hydrogen in plastics to effectively neutralize the oxygen in biomass, thereby promoting the decomposition of biomass. At the same time, the oxygen-free radicals in biomass help

accelerate the degradation of plastics [54]. In the SHAP value analysis, RT ranks fourth, which is consistent with the findings of Timilsina et al. [55], who concluded that the heating rate has a significant impact on bio-oil yield. A higher HR value indicates a shorter time available for secondary reactions, thereby enhancing the bio-oil yield [56]. The impact of various input features has changed, which is likely due to the DNN's ability to capture deeper associations between input features after adding the three-component data [5]. This is reflected in the noticeable changes in the importance and ranking of input features in the SHAP value analysis, with Hp's contribution to the predictive accuracy of bio-oil yield being highlighted.

4. Conclusion

In conclusion, this study innovatively introduces three-component data into the prediction of bio-oil yield from biomass and plastic co-pyrolysis and establishes machine learning models based on LightGBM and DNN algorithms. The inclusion of the three-component data leads to significant improvements in model accuracy and performance, with the DNN model outperforming the LightGBM model; specifically, the R^2 of the DNN model increased from 0.817 to 0.931, while the MAE decreased from 4.655 to 3.583 and the RMSE reduced from 6.220 to 4.573. Combined SHAP value and PCC analyses indicate that the importance of features has changed after incorporating the three-component data, with the PR and Hp identified as key influencing factors in the predictive model. This research provides new insights into the coupling of machine learning and biomass co-pyrolysis, aiming to enhance the quality and yield of bio-oil and promote the application and development of renewable energy. In future research, other machine learning models, such as XGBoost and SVM, will be employed to conduct a more in-depth study on the incorporation of the three components of biomass. Experimental validation will also be integrated to explore their further impacts.

CRedit authorship contribution statement

Chenxi Zhao: Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Qi Xia:** Writing – original draft, Software, Data curation. **Siyu Wang:** Supervision, Data curation. **Xueying Lu:** Writing – review & editing. **Wenjing Yue:** Visualization. **Aihui Chen:** Writing – review & editing. **Juhui Chen:** Resources.

Data availability

Xia, Qi (2024), "Bio-oil yield from co-pyrolysis of biomass and plastic", Mendeley Data, V1, <https://doi.org/10.17632/xyn9bg36wr.1>.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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